

Render Environment Variables Reference

Comprehensive list of every environment variable the platform reads, what it controls, the default value, and when to flip it. Grouped by category for fast operator lookup.

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Quick reference - the only vars you usually touch

Variable	Stage	Default	When to change
FOXIFY_V2_ENABLED	Always	false	Set to true to mount routes (one-time)
SHADOW_AUTO_ACTIVATE	Shadow	false	Set to true to run signal-driven shadow loop
SHADOW_AUTO_POLICY	Shadow	conservative	Set to opportunistic or hybrid to test alternative policies
SS_TWO_SIDED_LIVE_ENABLED	LIVE money	false	Flip to true only when ready to ship real money
FOXIFY_V2_LIVE_EXECUTION	LIVE money	false	Must be true alongside SS_TWO_SIDED_LIVE_ENABLED
SS_TWO_SIDED_BOOT_HALT	Safety	true	Leave true; means every deploy starts halted until operator clears

Everything else has sensible defaults you usually don't need to think about.

1. Critical infrastructure (must be set for platform to function)

Variable	Type	Required?	Description
<code>POSTGRES_URL</code> or <code>DATABASE_URL</code>	URL	YES	Postgres connection string. Pair/event/audit persistence. Without this the service won't boot.
<code>PORT</code> or <code>API_PORT</code>	int	NO (default 4100)	HTTP port the API listens on. Render auto-sets <code>PORT</code> .
<code>HOST</code>	string	NO (default 0.0.0.0)	Host to bind.
<code>PILOT_ADMIN_TOKEN</code>	string (32+ chars)	YES	All <code>/admin/foxify/v2/*</code> endpoints require <code>X-Admin-Token</code> matching this value.
<code>FOXIFY_API_KEY</code>	string (32+ chars)	YES (for Foxify integration)	All <code>/foxify/v2/*</code> endpoints require <code>X-Foxify-Token</code> matching this value. Share with Foxify when they integrate.

2. Venue credentials (price feeds + execution)

Deribit

Variable	Type	Required?	Description
<code>DERIBIT_CLIENT_ID</code>	string	YES	Deribit API client ID. Required for live option quotes and live execution.
<code>DERIBIT_CLIENT_SECRET</code>	string	YES	Deribit API client secret.
<code>DERIBIT_ENV</code>	<code>live</code> or <code>testnet</code>	NO (default <code>live</code>)	Which Deribit environment to hit.
<code>DERIBIT_PAPER</code>	bool	NO	If <code>true</code> , uses Deribit paper trading.

Bullish

Variable	Type	Required?	Description
PILOT_BULLISH_ENABLED	bool	YES (set to <code>true</code>)	Enables Bullish as a venue for quotes/routing.
PILOT_BULLISH_AUTH_MODE	<code>ecdsa</code> or <code>hmac</code>	YES (set to <code>ecdsa</code>)	Authentication mode. Production uses ECDSA.
PILOT_BULLISH_ECDSA_PUBLIC_KEY	string	YES	Bullish ECDSA public key.
PILOT_BULLISH_ECDSA_PRIVATE_KEY	string	YES	Bullish ECDSA private key. Sensitive - rotate periodically.
PILOT_BULLISH_ECDSA_METADATA	string	YES	Bullish ECDSA metadata field.
PILOT_BULLISH_TRADING_ACCOUNT_ID	string	YES	Trading account ID for live execution.
PILOT_BULLISH_REST_BASE_URL	URL	NO (sensible default)	Override Bullish REST URL.
PILOT_BULLISH_PUBLIC_WS_URL	URL	NO (sensible default)	Override Bullish public WS URL.
PILOT_BULLISH_PRIVATE_WS_URL	URL	NO (sensible default)	Override Bullish private WS URL.
PILOT_BULLISH_DEFAULT_SYMBOL	string	NO (default <code>BTCUSDC</code>)	Default trading symbol.

3. Foxify v2 feature flags

Variable	Type	Default	Description
FOXIFY_V2_ENABLED	bool	false	Master switch to mount <code>/foxify/v2/*</code> and <code>/admin/foxify/v2/*</code> routes. Must be true for any v2 functionality.
SS_TWO_SIDED_LIVE_ENABLED	bool	false	Master switch for REAL-MONEY execution. Stays <code>false</code> during shadow testing.
FOXIFY_V2_LIVE_EXECUTION	bool	false	Secondary gate inside v2 flow. Both this AND <code>SS_TWO_SIDED_LIVE_ENABLED</code> must be <code>true</code> for real-money to execute (defense in depth).
SS_TWO_SIDED_BOOT_HALT	bool	true	System auto-halts on every deploy boot. Operator clears with <code>POST /admin/foxify/v2/resume</code> . Belt-and-suspenders.
SS_TWO_SIDED_MAX_PAIRS_PER_DAY	int	2	Hard cap on real-money activations per day. Increase incrementally as confidence grows.
SS_TWO_SIDED_CELL_ALLOWLIST	csv	pair_50k_2pct	Global cell allowlist (env-level). Per-regime allowlists are stored in DB; override via <code>/admin/foxify/v2/cell-allowlist</code> .
SS_TWO_SIDED_NEWBORN_REVIEW_PER_REGIME	int	10	After N newborn triggers in a regime, system auto-halts for operator review.

4. Shadow auto-activator (signal-driven shadow loop)

Variable	Type	Default	Description
<code>SHADOW_AUTO_ACTIVATE</code>	bool	<code>false</code>	Master switch. When <code>true</code> , background loop polls signal every <code>SHADOW_AUTO_POLL_MS</code> and fires shadow activations when policy permits. Zero real-money risk - all activations use <code>ShadowStrangleExecutor</code> .
<code>SHADOW_AUTO_POLICY</code>	<code>conservative</code> / <code>opportunistic</code> / <code>hybrid</code>	<code>conservative</code>	Determines which signal triggers activation: - conservative: only when global signal flips <code>good_to_activate=true</code> - opportunistic: when any cell shows verdict=PROFITABLE (regardless of global signal) - hybrid: either condition above
<code>SHADOW_AUTO_POLL_MS</code>	int (ms)	<code>60000</code>	How often the loop checks the signal.
<code>SHADOW_AUTO_SUSTAINED_SEC</code>	int (sec)	<code>60</code>	Global signal must be GO for this many seconds before conservative/hybrid activates (filters one-tick noise). Bypassed by pure-opportunistic activations.
<code>SHADOW_AUTO_MAX_PER_WINDOW</code>	int	<code>3</code>	Max activations per rolling window. Prevents activation spam.
<code>SHADOW_AUTO_MAX_CELL_TRIGGER_PCT</code>	float	<code>0.05</code>	Only cells with trigger band $\leq$ this % are eligible. Default <code>0.05</code> = 5%.
<code>SHADOW_AUTO_MAX_COST_USDC</code>	int	<code>100000</code>	Max shadow cost cap. (Irrelevant since shadow doesn't risk real capital, but harness for sanity check.)

5. Operational tuning (defaults are fine, override only if you need to)

Logging + observability

Variable	Default	Description
<code>TWO_SIDED_LOG_LEVEL</code>	<code>info</code>	Set to <code>debug</code> for verbose lifecycle logs.
<code>PILOT_TRUST_PROXY</code>	(unset)	Set to <code>true</code> if running behind a reverse proxy (Render uses one).

Circuit breaker (pilot-level, for the broader system)

Variable	Default	Description
<code>PILOT_CIRCUIT_BREAKER_MAX_LOSS_PCT</code>	<code>0.5</code>	Max session loss as % before circuit trips.
<code>PILOT_CIRCUIT_BREAKER_WINDOW_MS</code>	<code>86400000</code> (24h)	Loss measurement window.
<code>PILOT_CIRCUIT_BREAKER_COOLDOWN_MS</code>	<code>14400000</code> (4h)	Time to wait after trip before recovery.
<code>PILOT_CIRCUIT_BREAKER_MIN_SAMPLES</code>	<code>4</code>	Min trades before circuit can trip.
<code>PILOT_CIRCUIT_BREAKER_ENFORCE</code>	<code>true</code>	Set to <code>false</code> to disable enforcement (logging only).

Volume cover (older subsystem, runs alongside)

Variable	Default	Description
<code>VOLUME_COVER_ENABLED</code>	<code>false</code>	Enable the older volume-cover subsystem. Independent of foxify v2.
<code>VOLUME_COVER_TRIGGER_DETECTOR_ENABLED</code>	<code>true</code>	Volume-cover trigger detector poll.
<code>VOLUME_COVER_HEDGE_MANAGER_ENABLED</code>	<code>true</code>	Volume-cover hedge manager.
<code>VOLUME_COVER_AUTH_DISABLED</code>	<code>false</code>	Disable auth on volume-cover routes (dev only).

6. Required env-var checklist by deployment stage

Stage 1: Service mounted, no auto-activity (read-only state)

```
POSTGRES_URL
PILOT_ADMIN_TOKEN
FOXIFY_V2_ENABLED=true
DERIBIT_CLIENT_ID
DERIBIT_CLIENT_SECRET
PILOT_BULLISH_ENABLED=true
PILOT_BULLISH_AUTH_MODE=ecdsa
PILOT_BULLISH_ECDSA_PUBLIC_KEY
PILOT_BULLISH_ECDSA_PRIVATE_KEY
PILOT_BULLISH_ECDSA_METADATA
PILOT_BULLISH_TRADING_ACCOUNT_ID
```

Verify with: `curl -H "X-Admin-Token: $RENDER_ADMIN_TOKEN" .../admin/foxify/v2/diagnostics | jq '.feed.health, .liquidChainCache.venue_status'`

Expected: `"healthy"` and both venues `ok: true`.

Stage 2: Add shadow auto-loop + Foxify-facing endpoints

Add to Stage 1:

```
SHADOW_AUTO_ACTIVATE=true
SHADOW_AUTO_POLICY=conservative # or opportunistic / hybrid
FOXIFY_API_KEY=<generate via: openssl rand -hex 32>
```

Verify with: `curl -H "X-Admin-Token: $RENDER_ADMIN_TOKEN" .../admin/foxify/v2/shadow-auto/status | jq '.enabled, .config.policy'`

Expected: `true` and your chosen policy name.

Stage 3: Add real-money execution (only after extensive shadow soak)

Add to Stage 2:

```
SS_TWO_SIDED_LIVE_ENABLED=true
FOXIFY_V2_LIVE_EXECUTION=true
SS_TWO_SIDED_MAX_PAIRS_PER_DAY=2 # start tight, ramp up
SS_TWO_SIDED_CELL_ALLOWLIST=pair_50k_2pct,pair_50k_5pct_otm,pair_50k_4pct_otm_short
```

After deploy, clear boot halt:

```
curl -X POST -H "X-Admin-Token: $RENDER_ADMIN_TOKEN" .../admin/foxify/v2/resume \
-H "Content-Type: application/json" -d '{"kind": "atticus"}
```

7. Activation policy decision matrix

When choosing `SHADOW_AUTO_POLICY`:

Policy	Fires when...	Activation cadence	Use case
conservative	Global signal <code>good_to_activate=true</code> AND signal sustained ≥ 60 s	Low - depends on regime + VRP threshold crossings	"What if Foxify uses the strictest policy?" - validates the safest baseline
opportunistic	Any cell has verdict=PROFITABLE (regardless of global signal)	Higher - includes calm-regime cells with +EV	"What if Foxify uses the cell- level EV directly?" - validates the more aggressive policy
hybrid	EITHER conservative OR opportunistic conditions	Highest - broadest coverage	"What if Foxify uses both signals?" - validates the most- frequent policy

You can run all three policies in parallel by deploying with one, observing the audit log for a day, then switching - all decisions are tagged with `policy: <value>` in the `details` field.

Most operators start with `conservative` to validate the strict baseline, then move to `hybrid` for production after they're confident in the close stack.

8. Security notes

1. **Rotate `PILOT_ADMIN_TOKEN` and `FOXIFY_API_KEY` periodically.** Especially before going live with real money.
 2. **Never paste tokens in chat/issues/PRs.** Render dashboard is the only place they should live (besides your operator notes).
 3. **`PILOT_BULLISH_ECDSA_PRIVATE_KEY` is the most sensitive value.** Compromise = full account access. Rotate immediately if exposed.
 4. **`DERIBIT_CLIENT_SECRET` likewise.** Different account but same severity.
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9. Where to find variables in code

- `services/api/src/server.ts` - top-level platform flags
- `services/api/src/singleSide/twoSided/shadowAutoActivator.ts` - shadow loop config (`readAutoActivatorConfig`)
- `services/api/src/singleSide/twoSided/routes.ts` - auth tokens, exposed in `/admin/foxify/v2/diagnostics` `env` field
- `services/api/src/pilot/config.ts` - Bullish + Deribit credentials

`grep -rn "process.env" services/api/src/` will list every env reference if you need to audit comprehensively.

10. Verification snippets

After any env change + redeploy, run these to confirm correct state:

```
# Platform health + env reflection
curl -sS -H "X-Admin-Token: $RENDER_ADMIN_TOKEN" \
  "<api-base-url>/admin/foxify/v2/diagnostics" \
  | jq '{env, feed_health: .feed.health, venues: .liquidChainCache.venue_status}'

# Shadow auto-loop config
curl -sS -H "X-Admin-Token: $RENDER_ADMIN_TOKEN" \
  "<api-base-url>/admin/foxify/v2/shadow-auto/status" \
  | jq '{enabled, policy: .config.policy, totals}'

# Foxify token works (use the value of FOXIFY_API_KEY)
curl -sS -H "X-Foxify-Token: <FOXIFY_API_KEY>" \
  "<api-base-url>/foxify/v2/should_activate" \
  | jq '.signal_tier, (.cell_opportunities | length)'
```

All three should respond 200 with sensible data.